

MBAi 448 | Winter 2026

Generative AI

Today

- Assignment 5 / week 6 takeaways *15 minutes*
- What is likely and what is true? *20 minutes*
- Quick break *5 minutes*
- Generative AI case studies *20 minutes*
- Assignment walkthrough *20 minutes*

Assignment 5 / week 6 takeaways

Generative AI

Week 6 takeaways (via ChatGPT)

➤ **Generative AI Is a Platform Shift, Not a Feature Upgrade**

The evolution from Transformers (2017) to GPT-4 and multimodal systems shows a rapid acceleration in capability (timeline on pp. 5–6). This is not incremental automation — it's a foundational shift in how work involving language, images, and reasoning gets done. Leaders should think in terms of business model redesign, not just productivity gains.

➤ **Prompt Design = Managerial Leverage**

The framework on page 8 (Instruction, Context, Input Data, Output Indicator) demonstrates that outputs are highly sensitive to structure. The in-class “Prompt Clinic” (pp. 43–45) shows how adding rules, format constraints, and guardrails dramatically improves reliability and compliance.

Managerial insight: Prompt engineering is less about clever wording and more about governance, risk mitigation, and repeatability.

Week 6 takeaways (via ChatGPT)

➤ **Multimodal AI Expands the Addressable Opportunity**

Pages 27–30 define multimodal AI as integrating multiple modalities (text, image, audio, etc.). This means AI systems can now reason across formats — not just generate text but analyze documents, interpret visuals, generate code, and combine them.

For businesses, this expands use cases from chatbots to workflow automation, diagnostics, design, media, and analytics.

➤ **AI = Representation Learning at Scale**

The slide “Data $\approx$ Representation” (p. 33) underscores a core idea: these systems transform data into high-dimensional representations (embeddings) that encode meaning.

Strategic implication: Competitive advantage increasingly depends on **who controls the data representations** — proprietary datasets, domain-specific fine-tuning, and feedback loops.

Week 6 takeaways (via ChatGPT)

➤ **Generative AI Is a 10x Productivity Tool — With Risk**

The “10x” slide (p. 24) and Text-to-Code examples (pp. 22–23) highlight significant productivity gains. However, the classroom exercise reveals hallucination risk and compliance challenges.

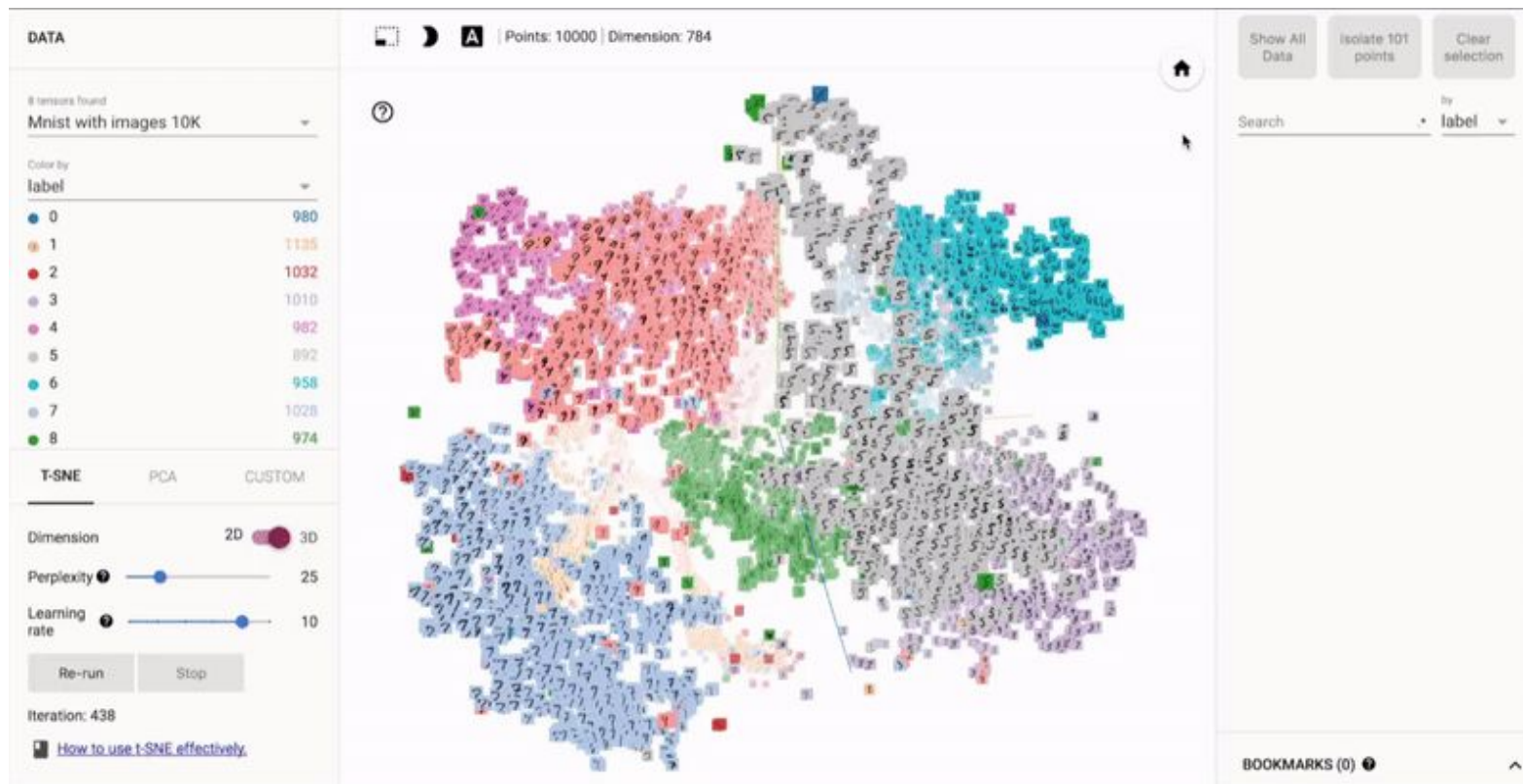
Executives must balance speed with oversight — instituting QA checks, governance, and risk controls before deploying AI in customer-facing contexts.

➤ **The Market Is Competitive and Rapidly Evolving**

The ecosystem slide (p. 7 & 31) shows intense competition (OpenAI, Anthropic, Google, DeepSeek, Mistral, etc.).

Implication for MBAs: Model capabilities will commoditize quickly. Durable advantage will come from integration, workflow redesign, proprietary data, and organizational change management — not from picking a single “best model.”

Deep learning builds latent representations

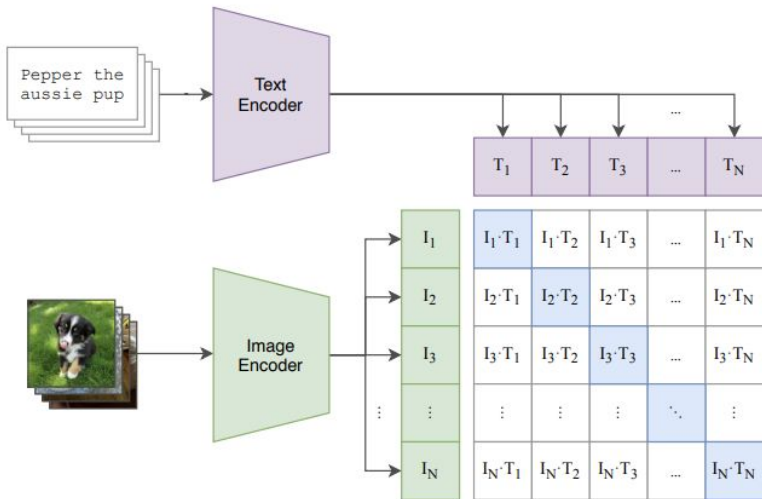


Different modalities must be aligned for use

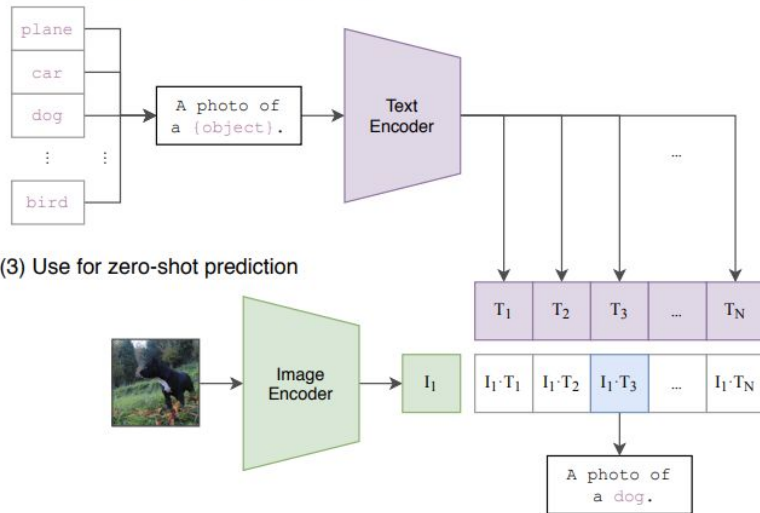
Learning Transferable Visual Models From Natural Language Supervision

2

(1) Contrastive pre-training



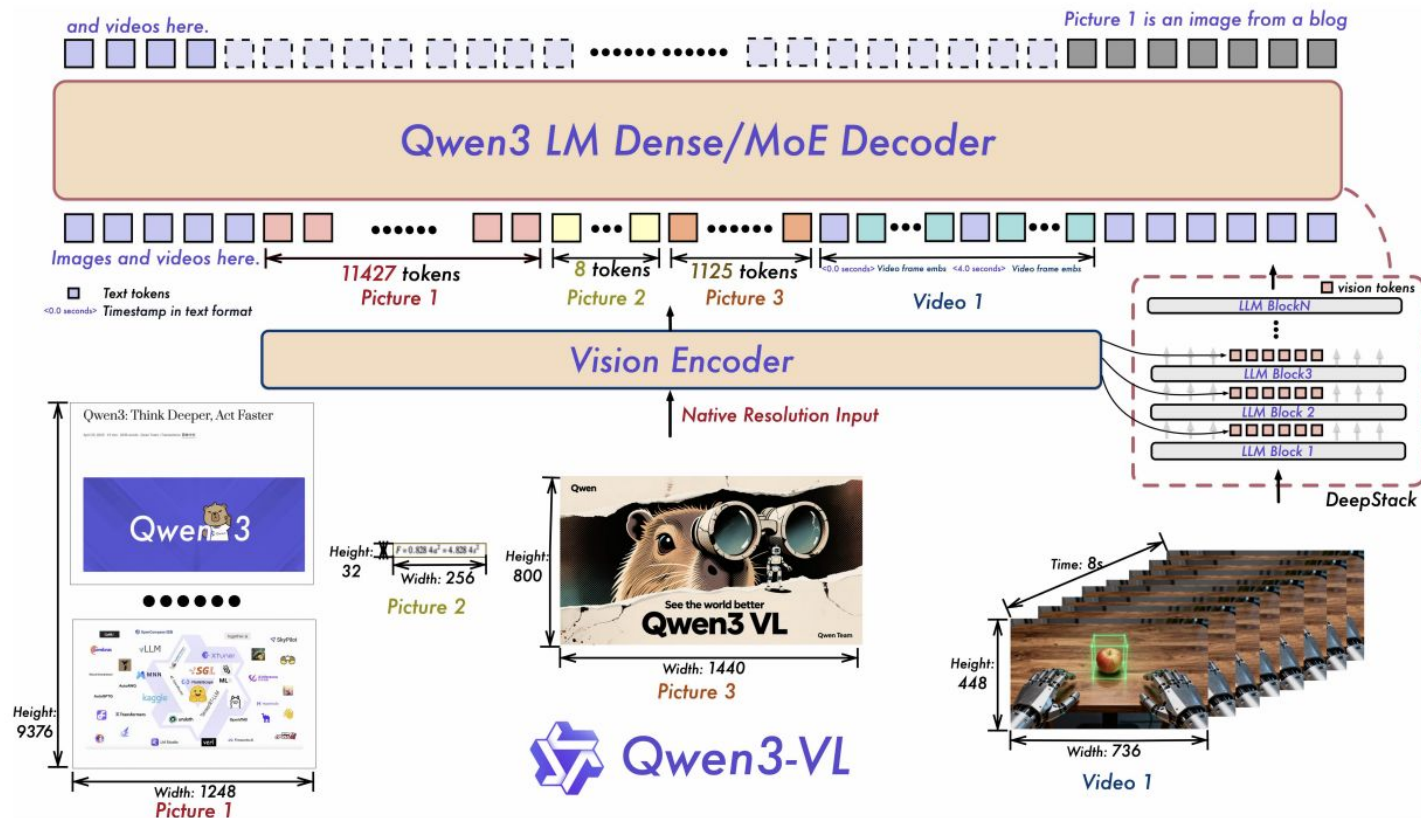
(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.

Different modalities must be aligned for use



Different modalities must be aligned for use

Table 3: Overview of training stages: data composition, token volumes, sequence lengths, and trainable components.

Stages	ViT Training	Joint Pre-training	Joint Long-context Mid-training
Data	Alt text Synthesis Caption Grounding, OCR, Video	+ Text, Knowledge Interleaving Video, OS Screenshot	+ High-quality Text & Multimodal Long Text, Long Video Reasoning, Long-CoT
Sequence length	4096	4096	32768→262144
Tokens	1T	15T	500B→200B
Training	ViT	ViT & LLM	ViT & LLM

parameters, utilizing 384 experts with 8 activated per token (sparsity of 48). For detailed descriptions of MuonClip, architecture design, and training infrastructure, we refer to the Kimi K2 technical report [53].

4.2 Model Architecture

The multimodal architecture of Kimi K2.5 consists of three components: a three-dimensional native-resolution vision encoder (MoonViT-3D), an MLP projector, and the Kimi K2 MoE language model, following the design principles established in Kimi-VL [54].

MoonViT-3D: Shared Embedding Space for Images and Videos In Kimi-VL, we employ MoonViT to natively process images at their original resolutions, eliminating the need for complex sub-image splitting and splicing operations. Initialized from SigLIP-SO-400M [77], MoonViT incorporates the patch packing strategy from NaViT [15], where single images are divided into patches, flattened, and sequentially concatenated into 1D sequences, thereby enabling efficient simultaneous training on images at varying resolutions.

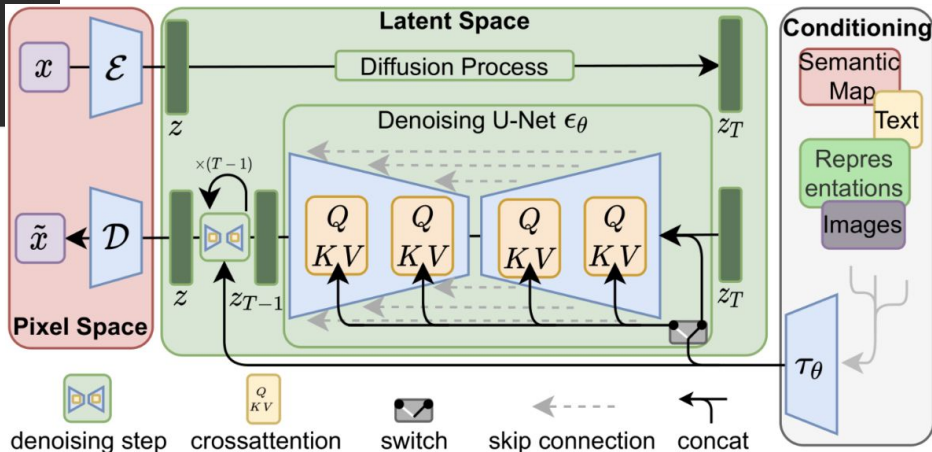
To maximize the transfer of image understanding capabilities to video, we introduce **MoonViT-3D** with a unified architecture, fully shared parameters, and a consistent embedding space. By generalizing the “patch n’ pack” philos-

Autoregression is not the only way to generate



Autoregression is not the only way to generate

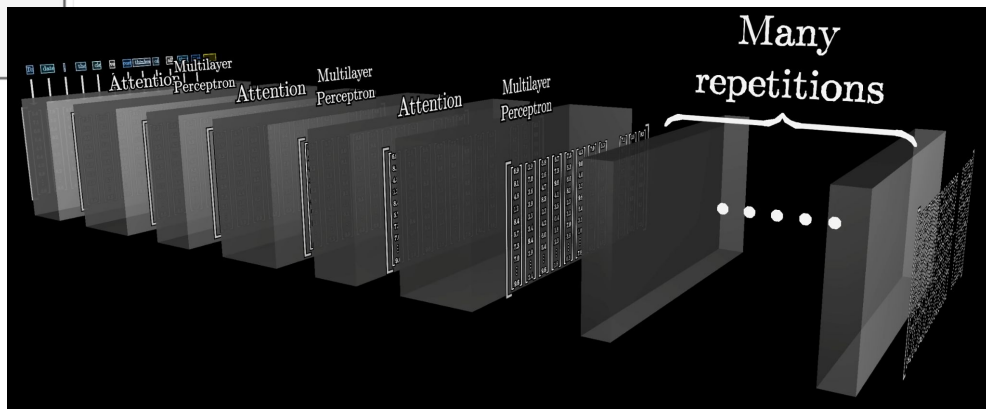
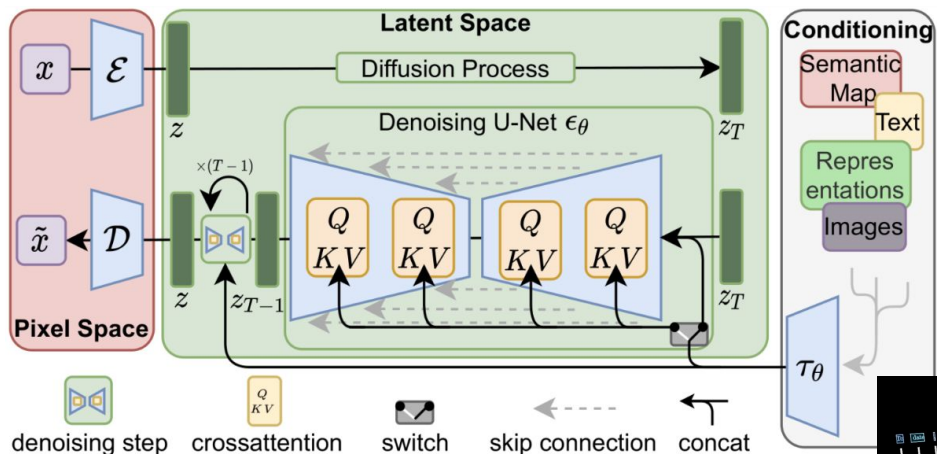
Image Generation by Reverse Diffusion (Denoising)



What is likely and what is true?

Generative AI

Generation is sampling from a learned space



Are you watching the Winter _____

First place receives a gold _____

The winning curling team was from _____

The most impressive event is the _____

The Winter Olympics happens every 4 _____

The 2050 Winter Olympics will be held in _____

2 + 2 equals _____

2026 ^ 2026 equals _____

Hallucinations can be undesirable

Mr. Schwartz said that he had consulted ChatGPT “to supplement” his own work and that, “in consultation” with it, found and cited the half-dozen nonexistent cases. He said ChatGPT had provided reassurances.

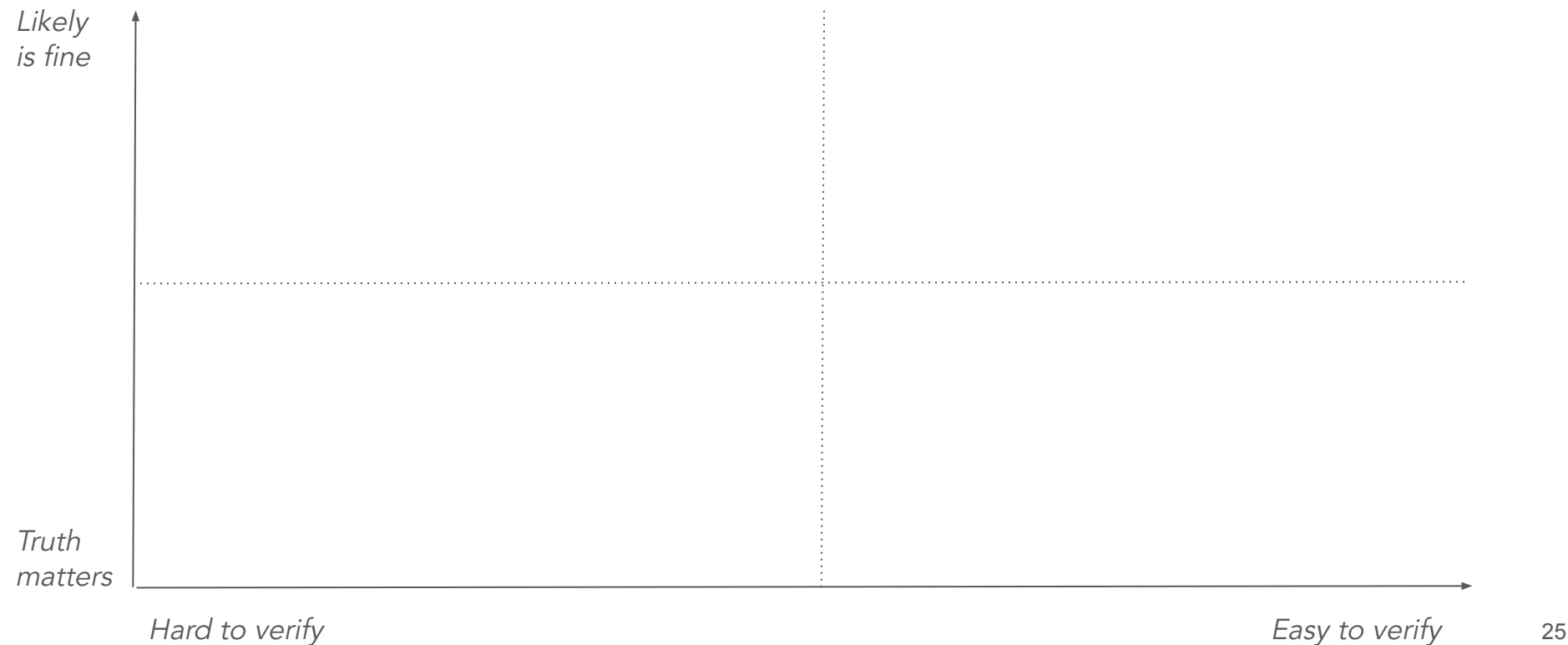
“Is varghese a real case,” he typed, according to a copy of the exchange that he submitted to the judge.

“Yes,” the chatbot replied, offering a citation and adding that it “is a real case.”

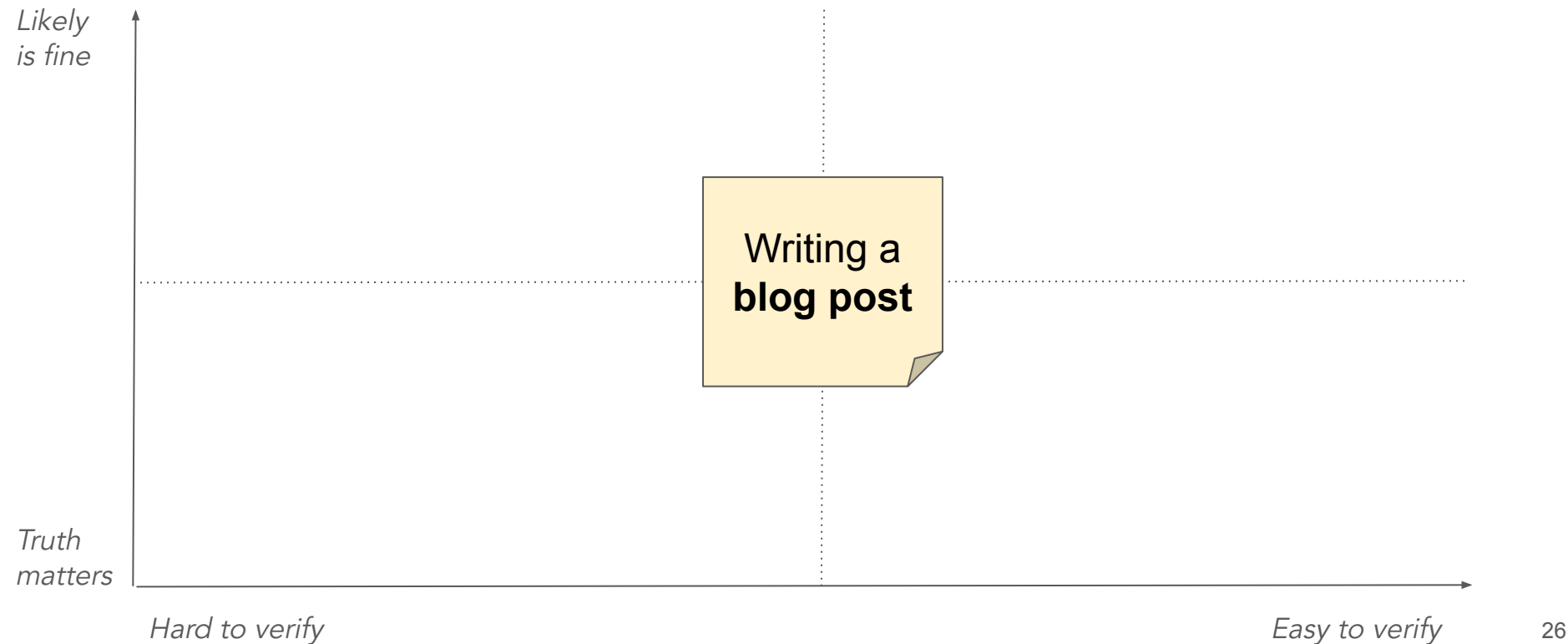
AI enables a variety of discrete capabilities

Capability	Task	Examples
Classify / Label	Assign inputs to predefined categories	Type, triage, diagnose, etc.
Extract	Identify a specific part of an unstructured input	Segmentation of image, audio, etc.
Group	Divide the input into distinct subsets	Cluster, detect anomalies, etc.
Score / Predict	Estimate a continuous numeric value	Forecast, quantify, rank, etc.
Search	Explore to find the best options for the input	Retrieve, route, match, etc.
Generate	Produce an output conditioned on an input	Draft, prototype, instantiate, etc.

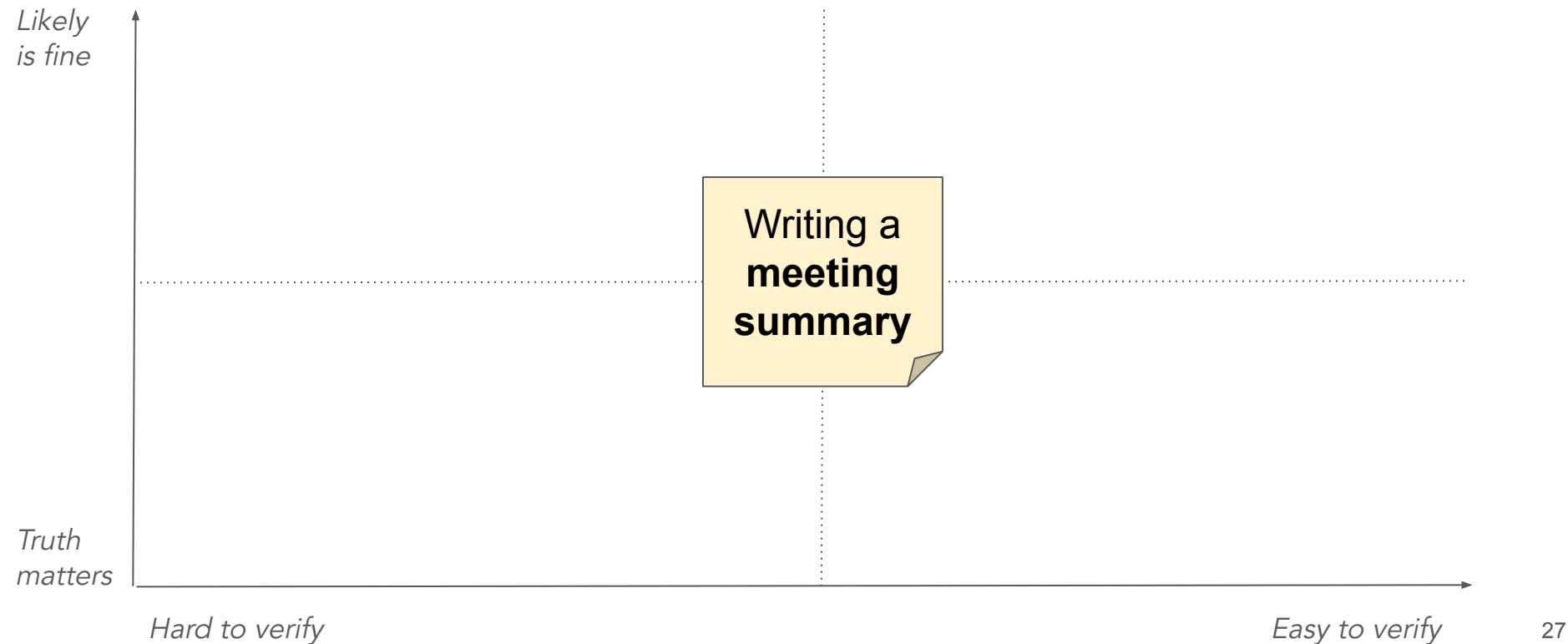
Sometimes likelihood is all you need



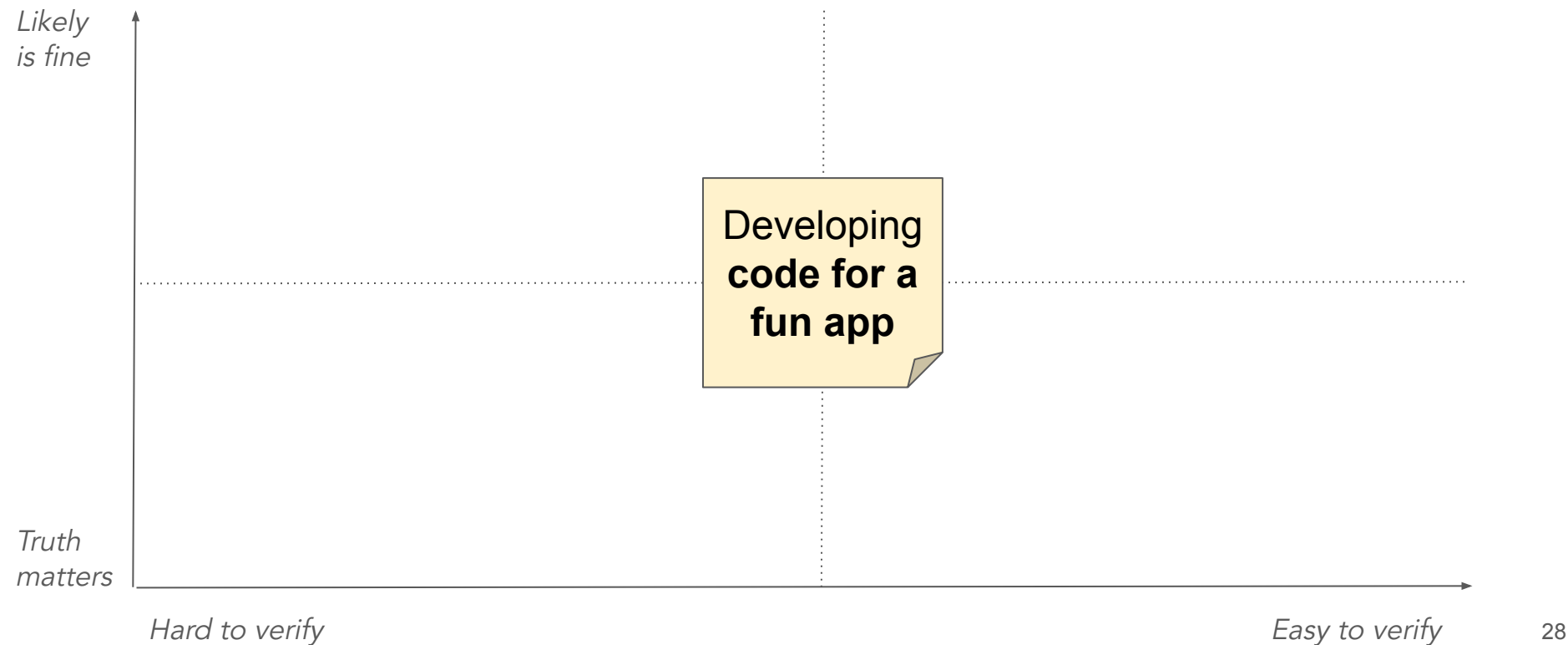
Sometimes likelihood is all you need



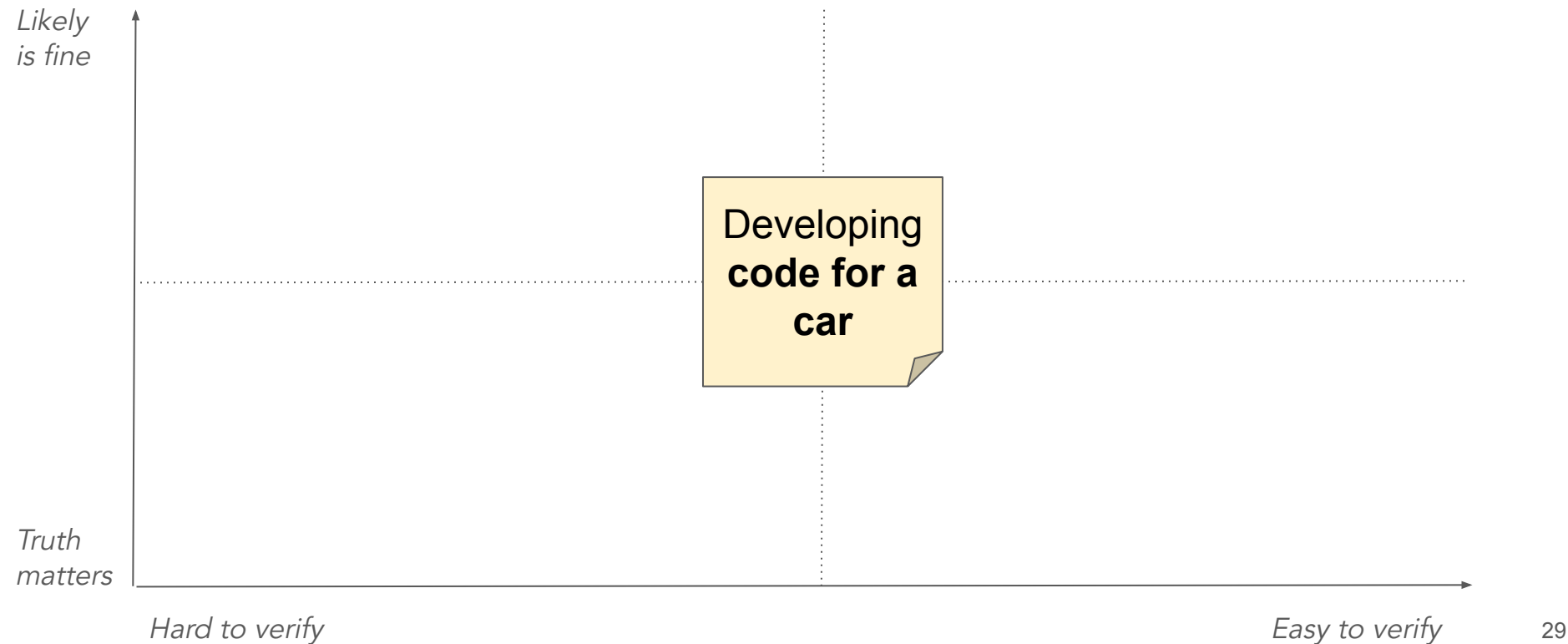
Sometimes likelihood is all you need



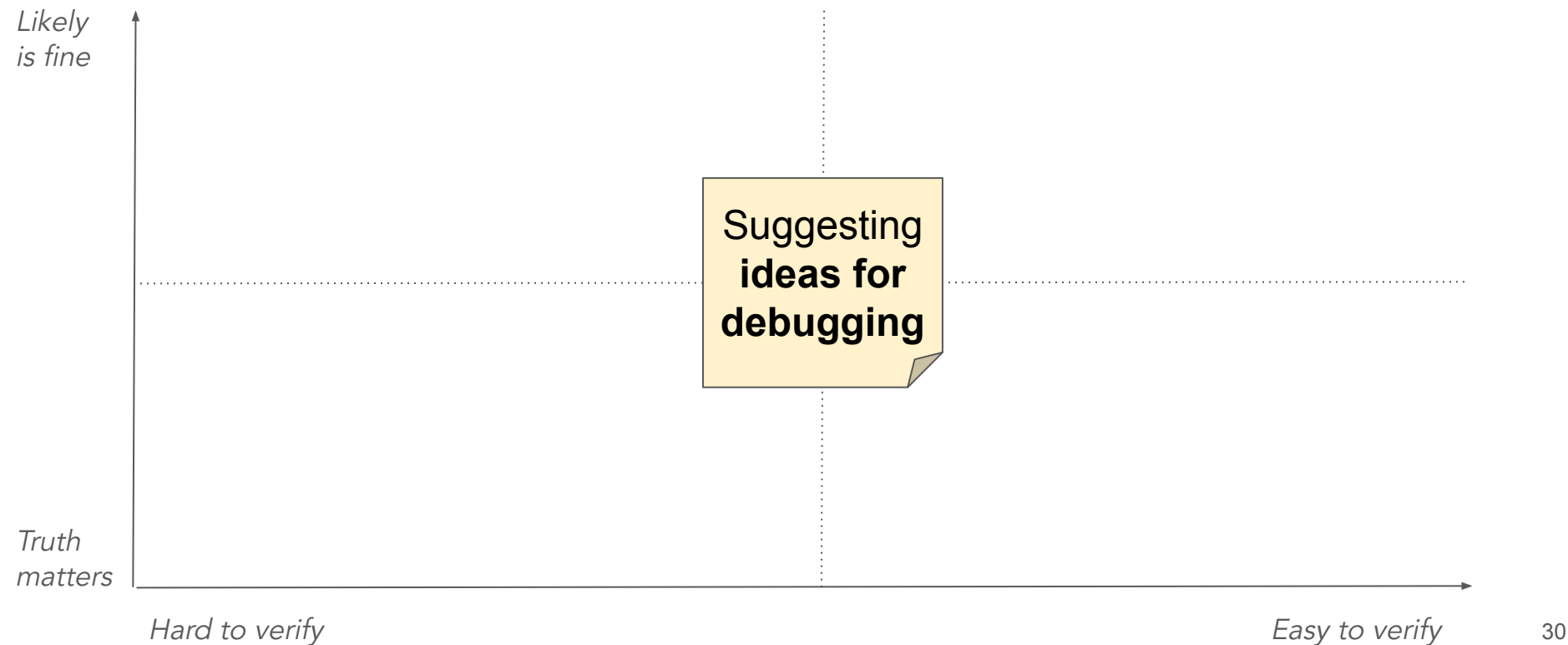
Sometimes likelihood is all you need



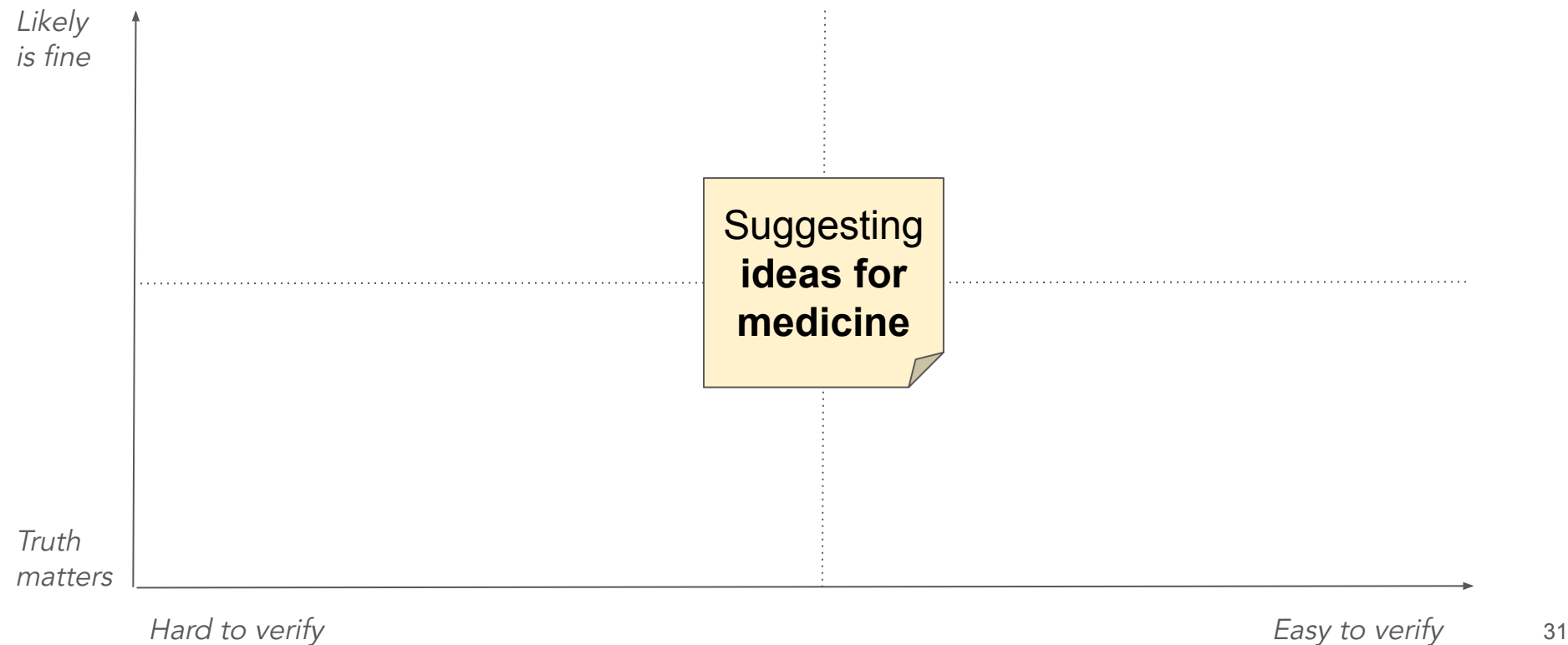
Sometimes likelihood is all you need



Sometimes likelihood is all you need



Sometimes likelihood is all you need

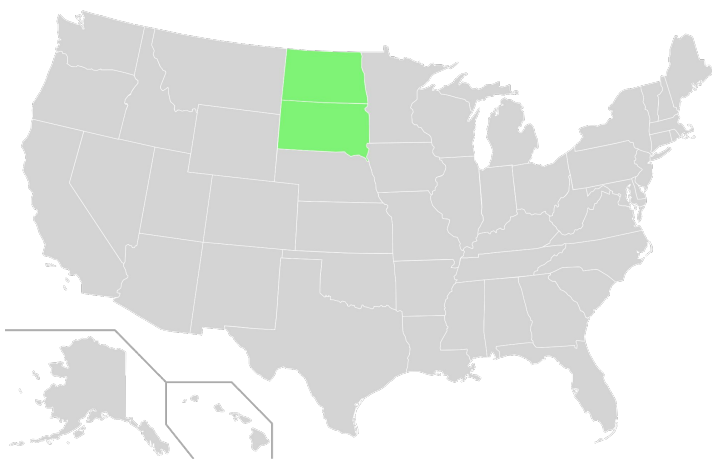


Prompting is all about steering likelihood



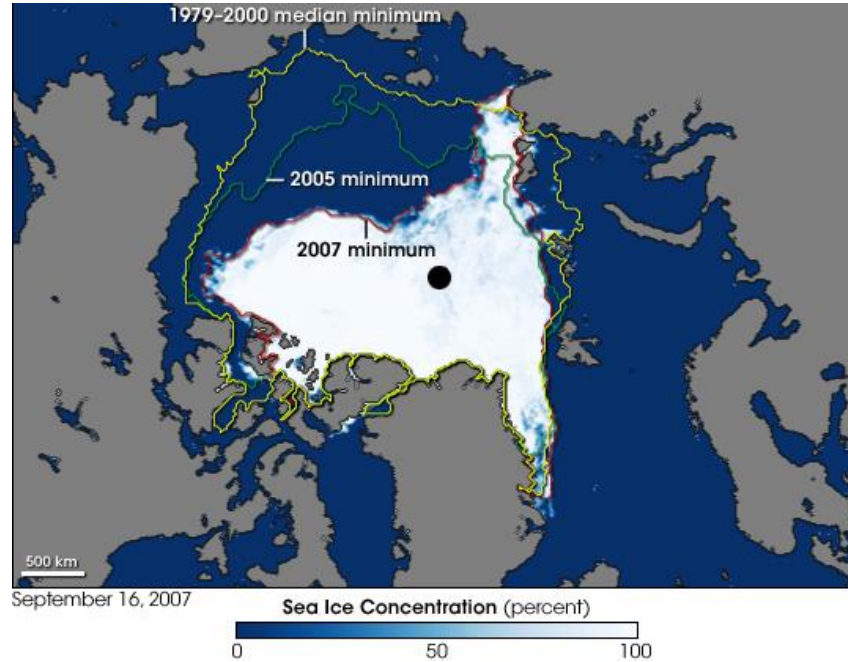
North and South _____

Prompting is all about steering likelihood



North and South _____

Prompting is all about steering likelihood



North and South _____

Prompting is not all you need



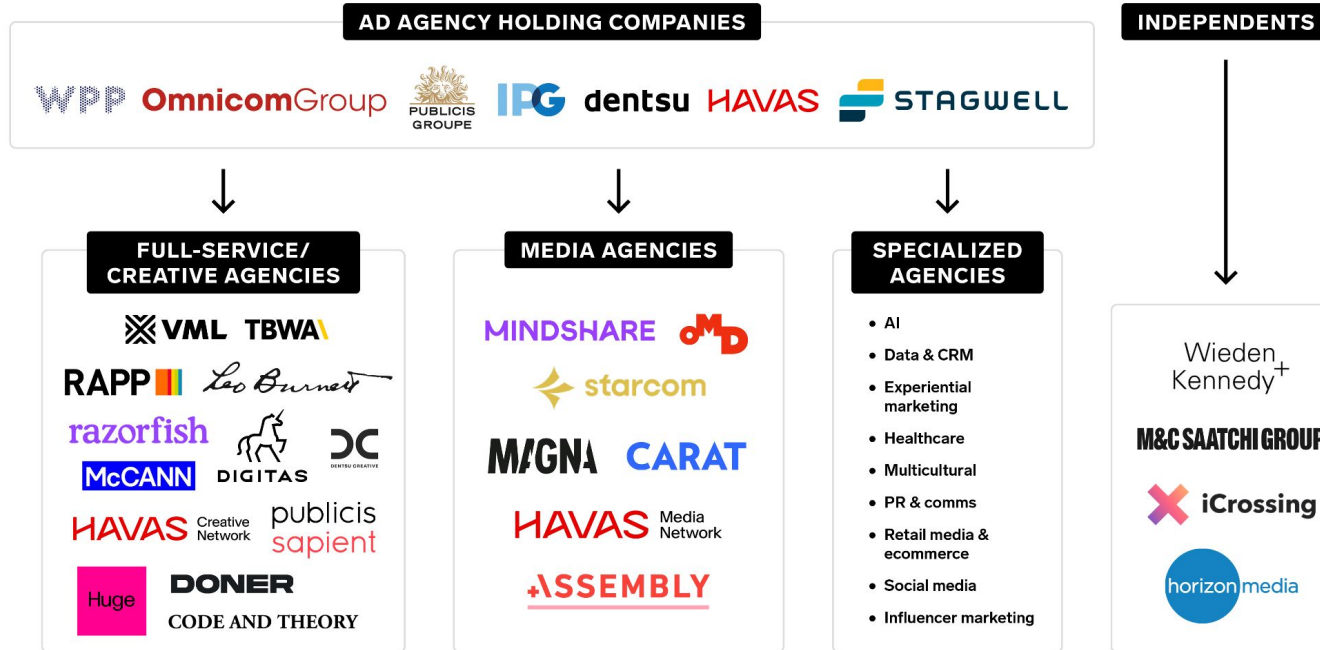
- The truth is important
- Verification is difficult
- Generation is hard
- The task requires judgement

Generative AI use cases

Generative AI

Case Study: Ad agencies

Ad Agency Holding Companies and Independents



Source: EMARKETER, Oct 2024

351594

Case Study: Ad age

Creative Agent

Image Generator

Video Generator

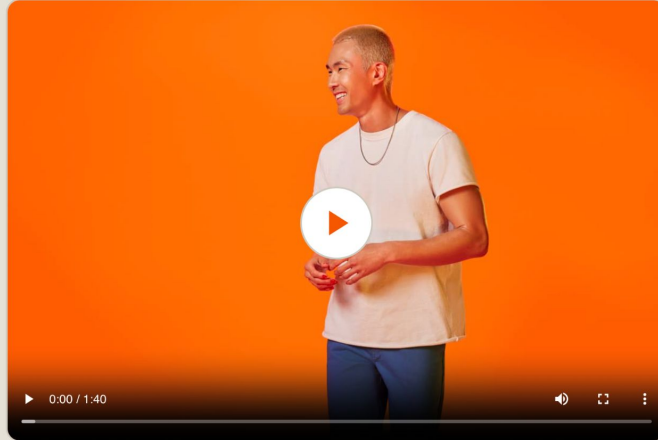
Audio Generator

Creative Studio

Creative Studio

Our centralized AI capabilities can help you self-serve all of your brand's creative needs within one experience.⁵

Get started with Creative Studio



Google Ads New campaign

- Campaign
- Ad group 1
- New ad 1
- Review campaign

New ad 1

Choose which type of ads to create

- Video ad Show ads with a single video
-
-

Ad name

Media assets

Videos



+ Add videos

Logos

Edit

Text assets

Headlines

+ Add headline 0/90

Long headlines

+ Add headline 0/90

Descriptions

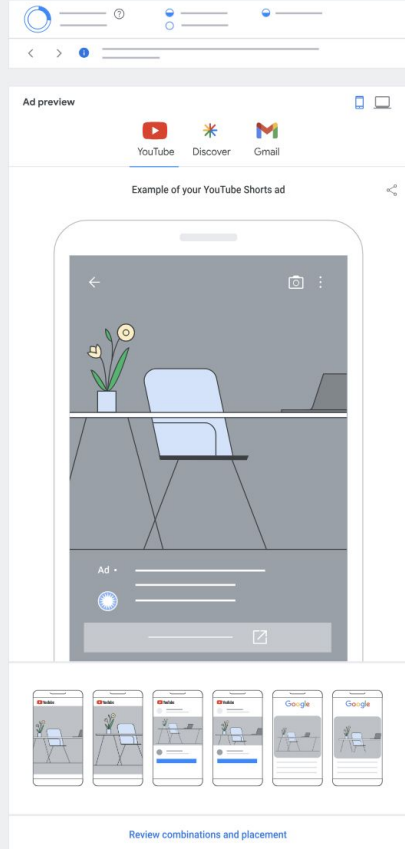
+ Add description 0/90

Call to action text

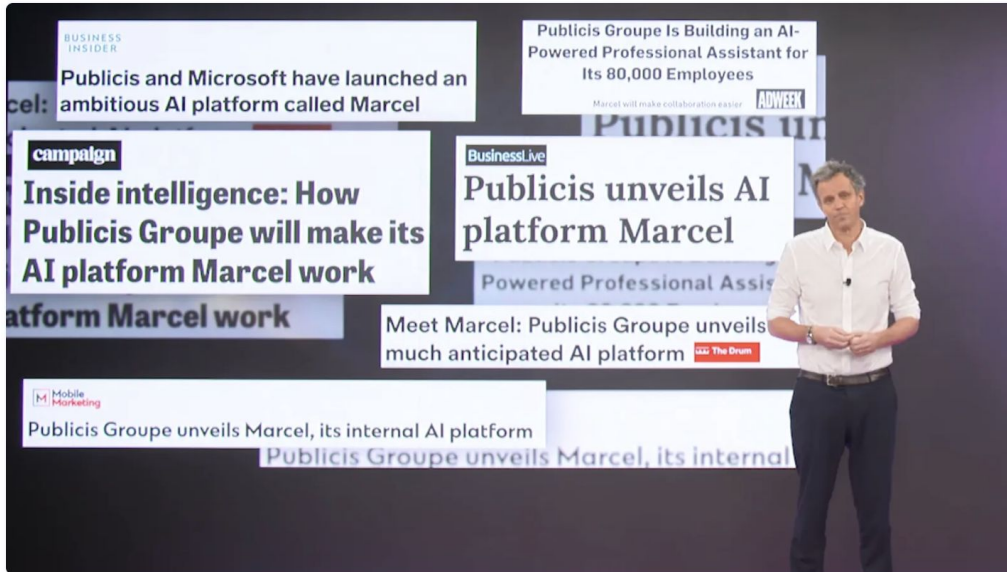
Business name

Sitelinks

+ Sitelink



Case Study: Ad agencies



From <https://www.marketingdive.com/news/publicis-generative-ai-CoreAI-future-of-agency-work/705581/>,
<https://www.wppopen.com/wpp-open-pro/overview>

WPP Open

Built for every marketer who wants to move at speed and scale

WPP Open Pro lets you



Plan – AI agents accessing WPP and partner data resources to identify opportunity



Create – the world's best GenAI and automation deployed to create on-brand, thumb-stopping ads



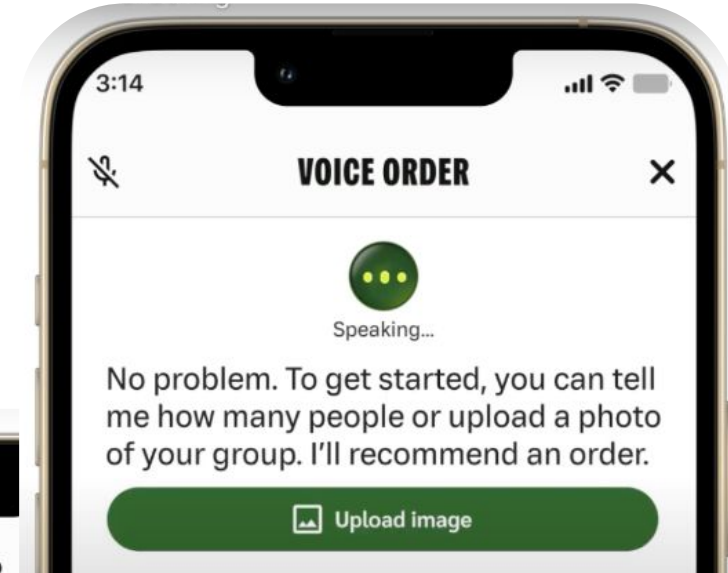
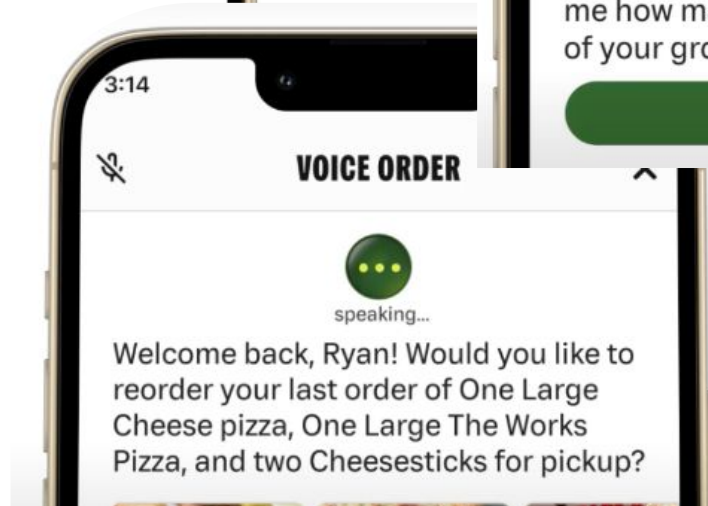
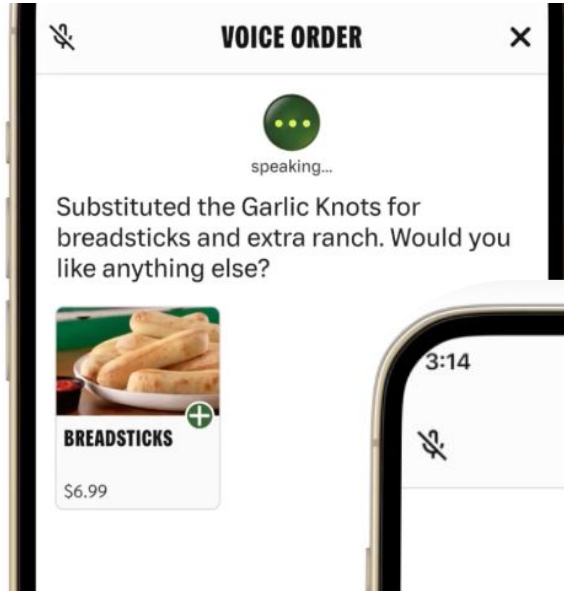
Publish – leverage the power of WPP Media or connect to major ad platforms to enable high-performing campaigns



All within one integrated platform

Case Study: Papa John's

Effortless
modifications



Case Study: Adobe



Case Study: Adobe

Create

Activate

Optimize

Brand intelligence

Generate more campaign content — faster.

Meet your content demands using AI to create and adapt on-brand ads and emails.

- Generate copy and images with AI models built for marketing teams.
- Reframe video ads automatically so they fit every format.
- Tailor content by persona, product, and region — including multilanguage creation and translation.
- Adapt and refine images for any placement using Generative Expand.
- Turn a campaign goal into all the creative assets you need using the Content Production Agent (coming soon).

[Learn about creation](#)

 Content Production Agent

Create 3 WKND Tent video ads targeting avid hikers to drive awareness.

 Campaign brief . PDF ✕

 Sure, extracting those messaging highlights now.



Case Study: Adobe

OpenAI | The
Walt Disney
Company



From <https://openai.com/index/disney-sora-agreement/> ,
<https://adage.com/technology/ai/aa-google-makes-ai-generated-ads-case-study-for-marketers/>

Google has released three fully AI-generated ads in the past two months. (Google)

Case Study: Replit

Still, turning that vision into a viable business of his own proved a lot harder than he anticipated. Replit was founded in 2016, and for eight long years, the company struggled to find product-market fit. “We had reached that \$2.83 million [in annual recurring revenue] back in ’21, maybe,” Masad recalled. “And so this is how painful it’s been. We’ve been hovering around the same revenue for like four or five years.”

The company tried selling to schools (“incredibly difficult,” Masad noted), cycling through different business models, and watched each one stabilize around the same modest revenue level.

Along the way, Replit built sophisticated infrastructure for cloud development environments and “multiplayer coding,” collaborative editing akin to Google Docs but for programming. But the technical achievement wasn’t translating into revenue growth, and by last year, with the company at 130 employees and burning through cash, Masad said he had to make a painful decision. “I

Case Study: Replit



2021 revenue:

- 2.83M

2023 revenue
(annualized):

- 150M

Case Study: Replit

AI Apps 50: Where Startups Spend on AI

1.  OpenAI	11.  Delve	21.  glean	31.  descript	41.  Otter.ai
2. ANTHROPIC	12.  perplexity	22.  Photoroom	32.  grammarly	42.  alma
3.  replit	13.  instantly	23.  Solve Intelligence	33.  manus	43.  applaud
4.  freepik	14.  customer.io	24.  GANMA	34.  Cognition	44.  CapCut
5. IIElevenLabs	15.  KlingAI	25.  clay	35.  OpusClip	45.  motion
6.  CURSOR	16.  Retell AI	26.  Cluely	36.  happyscribe	46.  crisp
7. Fyxr.ai 	17.  Canva	27.  crosby	37.  11x	47.  arcads
8.  Lorikeet	18.  Lovable	28.  Midjourney	38.  PLAUD	48.  emergent
9.  micro1.	19.  Metaview	29.  combinely	39.  Serval	49.  Read AI
10.  Notion	20.  ADEPT	30.  Merlin	40.  ada	50.  tavus

Some of the companies mentioned are portfolio companies of a16z. For a full list of portfolio companies, please visit a16z.com/investment-list.

Assignment walkthrough

Generative AI